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Comparison of Brain Structure Volumes in Insectivora and Primates.

III. Main olfactory bulb (MOB)

By

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With 5 Figures and 2 Tables

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Abstract and Summary: Volumes of the main olfactory bulb (MOB) in 76 species of Insectivora, Scandentia, Primates and Macroscelidea, and some of the laminar components in 34 species were measured. No statistically significant differences were found (1) between the two sides in the 162 individuals and (2) between males and females in the 19 species in which both sexes were examined. In interspecific comparisons the relative size (expressed by size indices) shows a definite tendency to decrease from Insectivora through prosimians and simians to man. The average indices were 100 – 64 – 10 – 3.6, respectively. Scandentia and Macroscelidea have the highest average MOB indices (135 and 166).

The relative MOB size is discussed in relation to feeding and social behaviour. It was shown that MOB development is largely linked to dietary adaptations but that its importance in feeding behaviour is paralleled by a similar importance in social behaviour. Within each dietary type, MOB development is associated with the different characteristics of the occupied niche. Among Insectivora, ground-dwelling species occupy the upper, and semiaquatic species the lower positions in the size scale for the MOB. Among Primates, in closely related species, the nocturnal species have in general better developed MOBs than the diurnal species.

The composition of the MOB is relatively stable, i.e. the laminar components (layers 1 + 2, 3, and 4–6) show no clear change in their percentage size from well developed to strongly reduced MOB. Only in the diurnal simians are layers 4–6 relatively small. This corresponds with the general observation (obtained from light-microscopy) that the granular layer (layer 6) is reduced and decomposed in higher Primates, and especially in man.

Introduction

The main olfactory bulb (MOB) is the primary brain center of the olfactory system and relay station for all olfactory impulses between olfactory mucosa and higher olfactory centers. Consequently, its size may be regarded as an important criterion in evaluating the significance of the main olfactory system of a given species. Details and a discussion about (1) the relationship between the size and structural differentiation and (2) the importance of including Insectivora in our comparative studies on primate brains were given in the Introduction of Part II of the present series dealing with the accessory olfactory bulb (STEPHAN et al., 1982). They are correspondingly valid also for MOB comparisons and, therefore, shall not be repeated here.

The MOB shows a general tendency to decrease in size from lower Insectivora through prosimians, mon-

keys, apes and finally man, as described by STEPHAN (1966, 1975), and STEPHAN and ANDY (1969, 1970). STEPHAN (1975) pointed out how clear the reduction is when the main olfactory system is not used in detecting food. This is not only valid for marine forms like seals and whales, but also in semiaquatic Insectivora and those Chiroptera which catch their prey in flight. Such species have clearly less developed MOB than purely terrestrial forms of Insectivora and fruit eating forms of Chiroptera (STEPHAN and PIRLOT, 1970; BHATNAGAR and KALLEN, 1974).

Besides detection and examination of food, other partial functions of the olfactory system may be related to general orientation in space, scent, detection of enemies, identification of related individuals and reproduction. In this presentation, we shall compare the volume size of the MOB and some of its components in a broad variety of species and fully discuss the results in relation to eco-ethological conditions.

Material and Methods

Volume size of the MOB was investigated in 162 individuals belonging to 76 species (Table 1). Volumes for left and right sides were measured separately and summed up, except for investigating intraindividual differences. Volumes of four main components were measured in 95 individuals of 34 species (Table 2, columns 2–6). The data were obtained from serial sections and were converted to fresh volumes. More details about the material sampled and on the method of measurements are given by STEPHAN et al. (1981a) in which body weights and the volumes of the total (net) brains and telencephali are also given. These data were used in this study to calculate size indices and percentage relations of the MOB (Table 1, columns 4–6).

In the measurements, the MOB included those parts of the olfactory nerves which are attached to the MOB and form its first layer. Within the MOB, not all of the layers (for structural pattern see STEPHAN, 1975) can be separated with the exactness required for separate measurements. This is possible only for four components: combined layers 1 + 2, 3, 4–6, and periventricular zone (PvZ) which includes the ependyma of the olfactory ventricle. The limit between the second and third layer is usually marked by the inner border of a band of external granular cells, the limit between the third and fourth layers by the generally sharp outer border of the mitral cell band, and the limit between sixth layer and the periventricular zone by the presence of granular cells in the former, which are absent in the latter.

In poorly developed MOBs, the separation of these main components can be fairly difficult. In such cases the data given for a single species may not be completely accurate. However, the trends obtained from the measurement of

a greater number of species may be expected to be fairly representative.

Interspecific comparisons were performed by evaluating the size differences of the MOB in isoponderous species by size indices (SI). In this method, the volumes of the structures under investigation are plotted against body weight in double logarithmic scales, and regression lines *within narrowly related groups* (i.e. in general, within families or subfamilies) are determined. The slopes of the regression lines indicate the rate at which the brains or the brain structures enlarge with increasing body weights. The method was described in more detail in Part I of this series dealing with the neocortex (FRAHM et al., 1982).

Slopes close to 0.57 were found in the MOB as well as in all structures of the main olfactory system and most structures of the limbic system. Thus, this value is applied in the present paper for the reference line of the MOB (Fig. 1). A reference line with this slope was plotted through the average log MOB volume and average log body weight of the 'basal' Insectivora. The distance of the plotted point for each species measured vertically above or below the reference line is expressed by an 'index of size difference' or 'size index' (SI). All the points on the reference line have indices of 1 or 100%. A doubling of the SI is reached at a distance of 0.301 (log of 2) from the reference line (see Fig. 1).

Since numerous authors still express interspecific size differences in percentage relative to larger complexes, we present lists of percentage values (Tables 1 and 2) complemented by several interstructural regression analyses in addition to the comparison by size indices. The pertinence of these methods in interspecific comparisons will be discussed below.

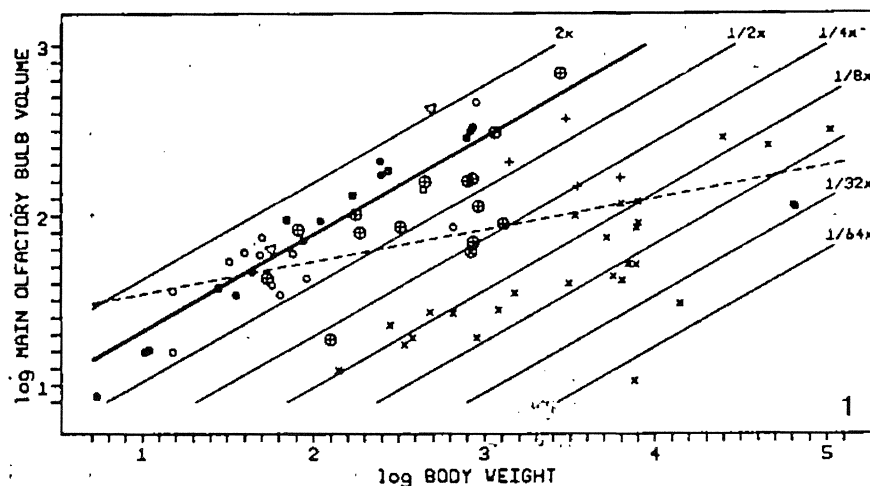


Fig. 1. Main olfactory bulb (MOB) volume plotted against body weight (BoW) in a double logarithmic scale. The volumes are in mm^3 , the weights in grams. The thick reference line is through the average MOB volume and average BoW of the 'basal' Insectivora (Tenrecinae, Erinaceidae, terrestrial Soricidae) with the common slope found for the families of Insectivora and Primates. The formula is: $\log \text{MOB} = 0.7526 + 0.57 \times \log \text{BoW}$. The thinner parallel lines are in multiples of 2 or of 1/2 and refer to the size indices ($\times 100$) given in Table 1, column 4 and scaled in Figures 2–5.

The stippled line is the overall regression line for the total of the 76 species, i.e. across the four orders represented in the material. The formula is $\log \text{MOB} = 1.3514 + 0.188 \times \log \text{BoW}$.

- 'basal' Insectivora, ○ 'progressive' Insectivora, ■ Scandentia (tree-shrews), + prosimians; × non-human simians;
- man; ◁ Macroscelidea. Nocturnal Primates are encircled.

Results

Intraindividual comparison

Variability: The MOBs of the two sides of an individual may vary to a great extent. In the 162 individuals, the standard error of the mean in % of the mean (SEM) was on the average of 4.0%. The maxima were found in individuals of *Callimico* (animal No. 6201; SEM 73.9%), *Cebuella* (animal No. 5923; SEM 37.1%), *Erythrocebus* (animal No. 1340; SEM 23.9%), *Nasalis* (animal No. 1365, SEM 21.9%), and *Gorilla* (animal No. A375; SEM 21.6%). However, in species with more than one individual (e.g. *Cebuella*, *Erythrocebus*) other individuals had distinctly lower SEM values. All the species mentioned have relatively small MOBs with size indices lower than 14 (see Table 1, column 4). Evidently, there is a certain concentration of high SEM values in species with poorly developed MOBs. The Insectivora, with their generally well developed MOBs, on the average have a clearly lower intraindividual variability (2.5%) than the simians, which generally have poorly developed MOBs (7.7%). However, high SEM values occasionally occur in species with well developed MOBs also, e.g. in an individual of *Erinaceus algirus* (animal No. 1192; SEM 13.0%; index 112). Thus, the MOBs of two sides within a brain can vary considerably and the question arises whether or not more generally the MOB of one of the two sides is significantly larger than that of the other.

Left-right distribution: In the 20 specimens of *Tupaia glis* the MOB was larger on the left side in 10 cases

and larger on the right side in 10 cases. No statistically significant size differences between the two sides were found. This, of course, does not rule out the possibility, that such differences of MOB size may exist in other species. However, our material on a single species was too limited to demonstrate significant differences. Even in *Sorex araneus*, where in 5 specimens the right side was larger in four cases, and in *Erinaceus europaeus*, where in 5 specimens the left side was larger in four cases, no statistically significant differences were found. In the total sample of the 162 individuals, without regard to species, the MOB was of equal size in one case, larger on the left side in 75 cases and larger on the right side in 86 cases. With regard to *Sorex* and *Erinaceus*, the slight numerical predominance of the right side in the overall material was insufficient to demonstrate any real difference as far as side is concerned. Our data generally suggest that differences in right and left sides do not exist.

Laminar components: The general statements on intraindividual variability and left-right distribution are valid not only for the total MOB, but also for its four measured components (layers 1 + 2, 3, 4–6, and PvZ). They will therefore not be discussed in greater detail. Data will be made available upon request.

Intraspecific comparison

Variability: Two or more individuals were measured in 48 species thus providing data on the intraspecific variability of the MOBs (both sides taken together).

Table 1. Data on total MOB (layers 1–6 + periventricular zone).

code number of species	species name	n	volume in mm ³	SEM in %	size index	MOB in % of	
						net brain	telen- cephalon
		(1)	(2)	(3)	(4)	(5)	(6)
Basal Insectivora							
0577	<i>Tenrec ecaudatus</i>	3	314	5.9	120	136	223
0579	<i>Setifer setosus</i>	2	209	1.8	160	149	231
0581	<i>Hemicentetes semispin.</i>	1	93.6	—	114	124	213
0585	<i>Echinops telfairi</i>	2*	71.2	11.5	98	126	212
0689	<i>Erinaceus algirus</i> ¹	3	285	15.2	112	100	161
0695	<i>Erinaceus europaeus</i>	5*	332	3.3	125	109	172
0701	<i>Hemiechinus auritus</i>	3	175	9.2	133	102	163
0715	<i>Sorex araneus</i>	5*	15.7	9.0	73	83	135
0769	<i>Sorex minutus</i>	1	8.65	—	59	84	142
0943	<i>Crocodyra flavescens</i> ²	2	37.6	6.2	99	92	156
1071	<i>Crocodyra russula</i>	2*	16.3	5.4	74	92	154
1137	<i>Suncus murinus</i>	1	34.2	—	79	97	164
Progressive Insectivora							
0575	<i>Solenodon paradoxus</i>	2*	464 +	4.0	170	108	179
0589	<i>Oryzomys talpoides</i>	1	46.9	—	96	87	141
0595	<i>Microgale cowani</i>	1	35.9	—	135	92	145

Table 1. (Continued)

code number of species	species name	n	volume in mm ³	SEM in %	size index	MOB in ‰ of	
						net brain	telen- cephalon
		(1)	(2)	(3)	(4)	(5)	(6)
0601	<i>Microgale dobsoni</i> ³	1	53.6	—	130	104	164
0629	<i>Microgale talazaci</i> ⁴	1	74.3	—	141	100	157
0633	<i>Limnogale mergulus</i>	1	42.9	—	58	41	83
0637	<i>Potamogale velox</i>	2	86.0	4.5	38	22	44
0639	<i>Micropotamogale lamottei</i>	3	34.1 ⁺	24.2	56	46	84
0643	<i>Chrysochloris asiatica</i>	1	58.5	—	112	89	136
0645	<i>Chrysochloris stuhlmanni</i> ⁵	2*	60.6	8.2	131	87	133
0839	<i>Neomys fodiens</i>	2	15.8	18.4	59	53	90
1205	<i>Desmana moschata</i>	1	141	—	78	39	66
1207	<i>Galemys pyrenaicus</i>	2	39.0	3.9	69	32	53
1215	<i>Talpa europaea</i>	2*	59.6	13.0	89	63	108
Scandentia							
1263	<i>Tupaia glis</i>	20*	131	8.2	125	44	70
1269	<i>Tupaia minor</i>	2	94.3	3.9	148	39	65
1289	<i>Urogale everetti</i>	1	184	—	133	46	76
Primates							
Prosimians							
3199	<i>Microcebus murinus</i>	6*	43.0	9.2	78	26	38
3203	<i>Cheirogaleus major</i>	2	158	5.7	86	25	37
3205	<i>Cheirogaleus medius</i>	2	102	7.3	94	34	53
3215	<i>Lemur albifrons</i> ⁶	3	207	8.1	59	9.3	13
3227	<i>Varecia variegata</i> ⁷	1	369	—	68	12	18
3239	<i>Lepilemur ruficaudatus</i>	3	113	20.5	41	16	24
3243	<i>Avahi laniger laniger</i>	2	89.2	7.4	27	9.1	14
3244	<i>Avahi l. occidentalis</i>	2*	70.1	16.7	26	7.7	12
3247	<i>Propithecus verreauxi</i>	2*	147	16.6	25	5.9	8.7
3249	<i>Indri indri</i>	2*	168	23.9	20	4.6	6.6
3251	<i>Daubentonia madagascar.</i>	1	685	—	131	16	23
3253	<i>Loris tardigradus</i>	1	85.8	—	56	14	19
3255	<i>Nycticebus coucang</i>	2	159	0.1	62	14	19
3259	<i>Perodicticus potto</i>	2	310	16.0	99	23	33
3265	<i>Galago senegalensis</i>	1	79.2	—	71	18	26
3267	<i>Otlemur crassicaudatus</i> ⁴	2	166	9.6	63	17	25
3275	<i>Galagoides demidoffi</i> ⁸	2*	83.1	5.2	120	26	37
3282	<i>Tarsius spp.</i>	2	18.8	12.4	21	5.5	8.3
Simians							
3287	<i>Callithrix jacchus</i>	3	22.6 ⁺	22.6	16	3.1	4.3
3289	<i>Cebuella pygmaea</i>	2	12.3	2.0	13	2.9	4.0
3306	<i>Saguinus midas</i> ¹⁰	2*	17.3	6.8	11	1.8	2.5
3311	<i>Saguinus oedipus</i>	3*	19.1	22.7	11	2.0	2.7
3315	<i>Callimico goeldii</i>	1	27.0	—	14	2.6	3.5
3324	<i>Cebus spp.</i>	2*	39.9	26.8	7.2	0.6	0.8
3325	<i>Aotus trivirgatus</i>	3	61.0 ⁺	18.1	23	3.8	5.0
3327	<i>Callicebus moloch</i>	2*	19.2	8.1	7.0	1.1	1.4
3335	<i>Saimiri sciureus</i>	1	26.8	—	12	1.2	1.5
3341	<i>Pithecia monachus</i>	2	34.8	11.2	9.5	1.1	1.4
3366	<i>Alouatta spp.</i>	2	41.4	14.9	5.0	0.8	1.1
3371	<i>Ateles geoffroyi</i>	1	90.4	—	9.5	0.9	1.1
3379	<i>Lagothrix lagothricha</i>	3	73.4	10.2	9.9	0.8	1.0
3391	<i>Macaca mulatta</i>	1	84.3	—	9.0	1.0	1.2
3407	<i>Cercopithecus albigena</i>	1	121	—	13	1.2	1.6
3415	<i>Papio anubis</i>	2*	287	5.5	16	1.5	1.9

Table 1. (Continued)

code number of species	species name	n	volume in mm ³	SEM in %	size index	MOB in ‰ of	
						net brain	telen- cephalon
		(1)	(2)	(3)	(4)	(5)	(6)
3433	<i>Cercopithecus ascanius</i>	1	99.5	—	17	1.6	1.9
3453	<i>Cercopithecus mitis</i>	1	117	—	14	1.7	2.1
3469	<i>Miopithecus talapoin</i> ¹¹	2*	27.8	25.4	8.6	0.7	0.9
3473	<i>Erythrocebus patas</i>	2	51.8	26.1	5.5	0.5	0.6
3477	<i>Colobus badius</i>	2	51.3	1.7	5.8	0.7	0.9
3493	<i>Pygathrix nemaeus</i>	1	10.7	—	1.2	0.15	0.2
3499	<i>Nasalis larvatus</i>	1	30.3	—	2.3	0.3	0.4
3539	<i>Hylobates lar</i>	1	43.9	—	5.6	0.45	0.6
3549	<i>Pan troglodytes</i>	1	257	—	10	0.7	0.8
3551	<i>Gorilla gorilla</i>	1	316	—	7.7	0.7	0.85
3553	<i>Homo sapiens</i>	1	114	—	3.6	0.09	0.1

Macroscelidea

7993	<i>Elephantulus fuscipes</i>	2	63.3	4.3	112	51	90
8011	<i>Rhynchocyon cirnei</i> ¹²	2	426	5.1	220	75	128

Means	range of species (code numbers)	N [†]					
Tenrecinae	0577—0585	3	4	6.4	123	134	220
Erinaceidae	0689—0701	3	3	9.2	123	104	165
Soricidae (terrest.)	0715—1137	3	5	6.9	77	90	150
Basal Insectivora		9	12	7.5	104	108	177
Oryzorictinae	0589—0633	0	5	—	112	85	138
Potamogalinae	0637—0639	2	2	14.4	47	34	64
Chrysochloridae	0643—0645	1	2	8.2	122	88	135
Talpidae	1205—1215	2	3	8.5	79	45	76
Progr. Insectivora		7	14	10.9	97	69	113
Insectivora	0575—1215	16	26	9.0	100	87	143
Scandentia	1263—1289	2	3	6.1	135	43	70
Cheirogaleidae	3199—3205	3	3	7.4	86	28	43
Lemuridae	3215—3239	2	3	14.3	56	12	18
Indriidae	3243—3249	4	4	16.2	25	6.8	10
Lorisidae	3253—3275	4	6	7.7	79	19	27
Prosimians	3199—3282	14	18	11.3	64	16	23
Callitrichidae	3287—3315	4	5	13.5	13	2.5	3.4
Cebidae	3324—3379	6	8	14.9	10	1.3	1.7
Cercopithecinae	3391—3473	3	7	19.0	12	1.2	1.5
Colobinae	3477—3499	1	3	1.7	3.1	0.4	0.5
Cercopithecidae	3391—3499	4	10	14.7	9.2	0.9	1.2
Pongidae	3539—3551	0	3	—	7.8	0.6	0.8
Simians	3287—3553	14	27	14.4	9.9	1.3	1.6
Primates	3299—3553	28	45	12.9	31	7.0	10
Macroscelidea	7993—8011	2	2	4.7	166	63	109

Footnotes to Table 1

Classification and species names were adapted to the list given by CORBET and HILL (1980). Names differing from those used in former papers and in STEPHAN et al. (1981a) are: 1 *Aethechinus algirus*, 2 *Crocidura occidentalis*, 3 *Nesogale dobsoni*, 4 *Nesogale talazaci*, 5 *Chlorotalpa stuhlmanni*, 6 *Lemur fulvus*, 7 *Lemur variegatus*, 8 *Galago crassicaudatus*, 9 *Galago demidovii*, 10 *Saguinus tamarin*, 11 *Cercopithecus talapoin*, 12 *Rhynchocyon stuhlmanni*.

n Number of individuals; 2 · n is number of measured MOBs.

* In these species both sexes are present.

+ Figures slightly different from those given by STEPHAN et al. (1981a) because of re-examination and/or additional data.

N Number of species.

† The first N refers to the number of species (with more than one animal investigated) which were used to calculate the figures in column 3. The second N refers to the number of species (with one or more individuals) which were included to calculate the figures given in column 4—6.

The average intraspecific standard error of the mean in % of the mean (Table 1, column 3) was 10.9%. The maxima were found in *Cebus* (SEM 26.8%), *Erythrocebus* (SEM 26.1%), *Miopithecus* (SEM 25.4%), *Micropotamogale* (SEM 24.2%), *Indri* (SEM 23.9%), *Saguinus oedipus* (SEM 22.7%), and *Callithrix* (SEM 22.6%). All these species have poorly or very poorly developed MOB's. The average SEM of the simian group was higher (14.4%, $n = 14$) than that of the prosimian group (11.3%, $n = 14$) and higher than the overall average (10.9%). Within the Insectivora, the average SEM was found to be clearly lower in the basal (7.5%, $n = 9$) than in the progressive forms (10.9%, $n = 7$), the latter of which include several species with reduced MOB's (Table 1, column 4).

Intersexual distribution: Although individuals of the two sexes were present in nineteen species examined (marked by an asterisk * in Table 1, column 1), in only four of them there was a sufficient number (five or more) of individuals for comparison. Of the twenty individuals of *Tupaia glis*, ten were male and ten female; of the six individuals of *Microcebus murinus*, one was male and five were female; of the five individuals of *Erinaceus europaeus*, three were male and one female (the sex of one animal was not known); and of the five individuals of *Sorex araneus*, three were male and two female. In all of these four species the females had slightly larger average MOB's than the males, but no statistically significant MOB size differences were found between males and females.

In the total sample of the nineteen species in which both sexes were present, the MOB was larger in males in seven species and larger in females in twelve species. With regard to the results of the four species (see above) which were examined statistically, the numerical predominance of MOB size in females was insufficient to demonstrate any real sex difference. Our data generally suggest that such differences do not exist. However, they do not of course rule out the possibility, that in individual species such differences in MOB size may exist.

Laminar components: Like the total MOB, the four measured components showed no significant sex difference in size.

Interspecific comparison of total MOB

MOB related to body weight (size indices): As described in the Materials and Methods section, a reference line with a slope of 0.57 was drawn through the average log MOB volume and the average log body weight of the twelve primitive (basal) Insectivora (Figure 1; thick line). The size differences of the MOB's in isoponderous species are expressed using the size indices (Table 1, column 4; and Figures 2–5).

The highest indices were found in Macroscelidea, Scandentia, Insectivora, and in some prosimians, the lowest in simians (Fig. 2). By far the highest index of the total sample (220) was found in *Rhynchocyon cirnei*, a representative of the Macroscelidea. In the second species, *Elephantulus fuscipes*, the index was only half this size (112), but still points to a well developed MOB. High indices are also found in all Scandentia (125–148).

In various families of the Insectivora high indices were found in the Solenodontidae, Erinaceidae, terrestrial Tenrecidae, and Chrysochloridae, and clearly lower indices were obtained in the Talpidae, Soricidae, and semiaquatic Tenrecidae (*Limnogale*, *Potamogalinae*) (Fig. 3). The size indices of the Insectivora range from 38 in *Potamogale* to 170 in *Solenodon*. Thus in *Solenodon* the relative size was about 4.5 times (170:38) larger than in an isoponderous *Potamogale*. Other high indices were found in *Setifer* (160) and *Microgale* spp. (130–141), other low indices were obtained in *Micropotamogale lamottei* (56), *Limnogale mergulus* (58), *Sorex minutus* and *Neomys fodiens* (59). These results indicate that clear differences in the relative size of the MOB already exist in Insectivora, i.e., not all Insectivora have very well developed MOB's. In several species, the low MOB development is clearly related to an adaptation to predatory life habits in fresh waters (marked W in Fig. 3; also see Discussion).

In the prosimians (Fig. 4), two species had indices above the average of the basal Insectivora (100). These were *Daubentonia madagascariensis* (131) and *Galagoides demidoff* (120). The lowest prosimian index was found in *Indri indri* (20). Thus, in *Daubentonia* the relative size of the MOB is about 6.5 times (131:20) larger than in an isoponderous *Indri*. Although the body weight of the indri was more than twice that of the aye-aye, its MOB was only one fourth the size (Table 1, column 2).

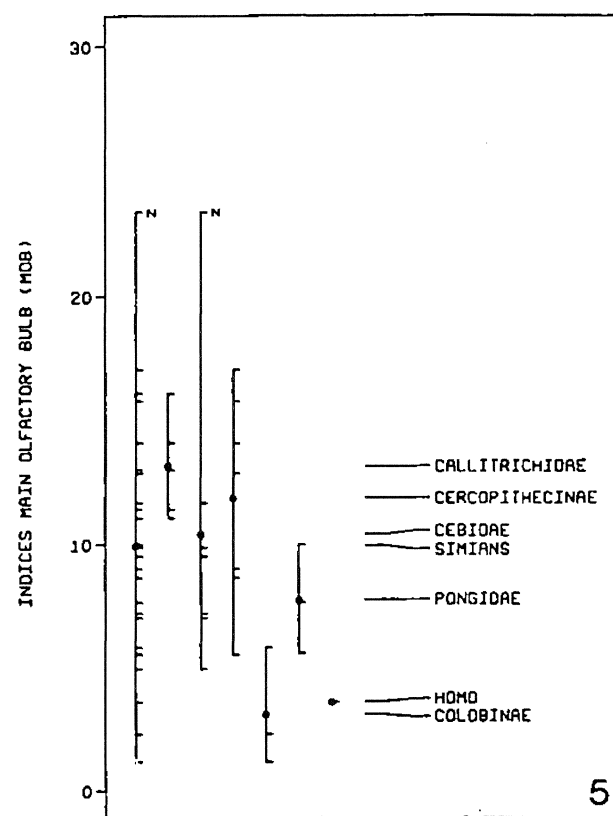
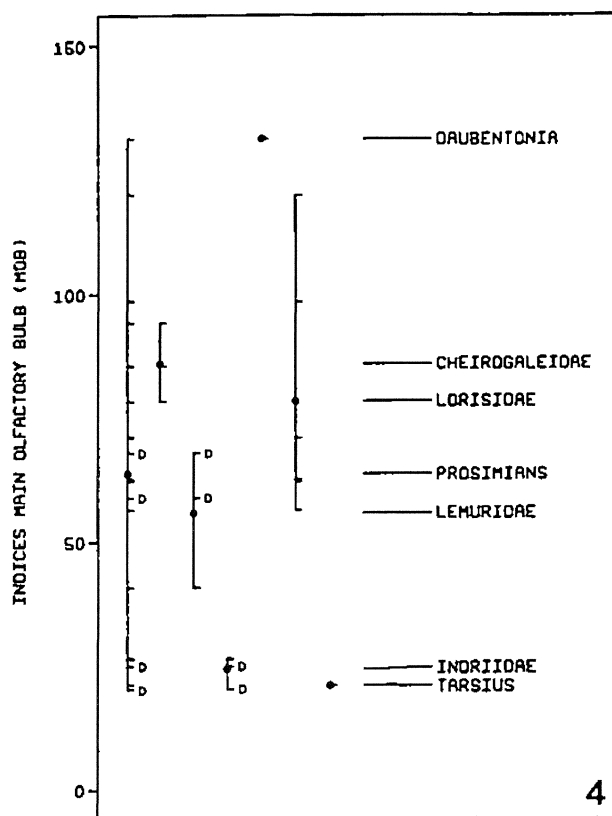
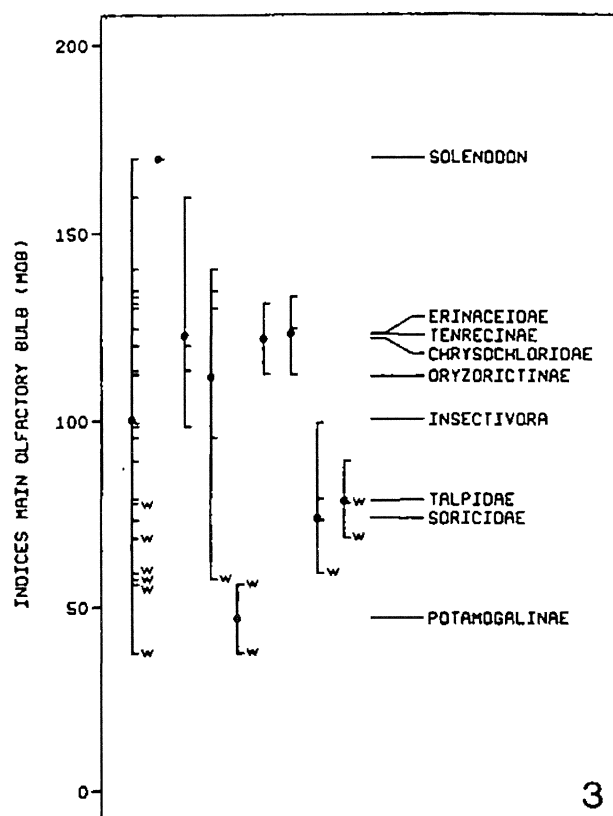
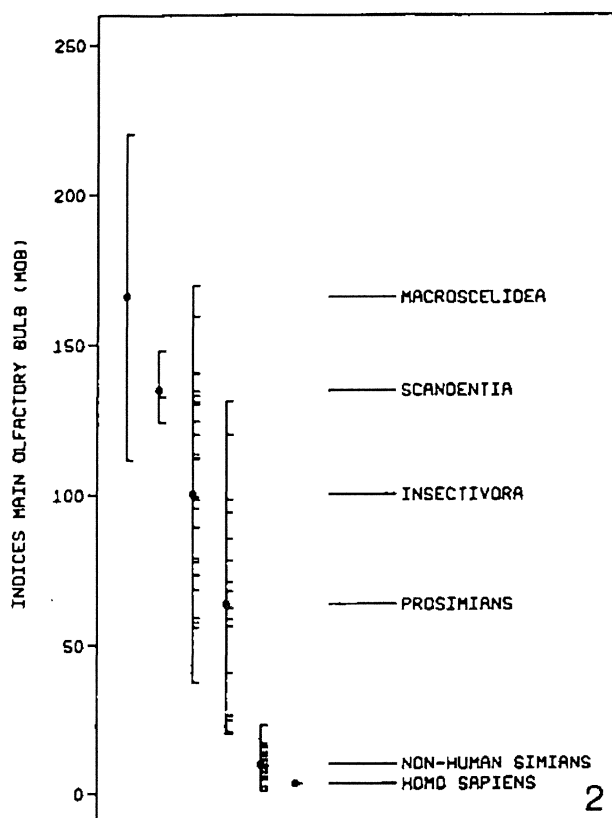
In various families or subfamilies of the prosimians, relatively high indices were found in the Cheirogaleidae (range 78–94; $\bar{x}$ 86; $n = 3$), Galaginae (range 63 to 120; $\bar{x}$ 85; $n = 3$) and Lorisinae (range 56–99; $\bar{x}$ 72; $n = 3$) in addition to *Daubentonia*; low indices were obtained in all the Indriidae (range 20–27; $\bar{x}$ 24.5; $n = 4$) and *Tarsius* (21) (Fig. 4).

In all simians (Fig. 5) the MOB was clearly reduced when compared with the average of the basal Insectivora. The highest index was found in *Aotus trivirgatus* (23), and the lowest in *Pygathrix nemaeus* (1.2), but in the latter species only one individual was available and its MOB appeared to be abnormally underdeveloped. The second minimum was found in *Nasalis larvatus* (2.3) and the third in *Homo sapiens* (3.6).

In various families or subfamilies of the simians, relatively high indices were found in the Callitrichidae

Table 2. Volumes and percentages of MOB components.

		n	Volumes in mm ³					Percentages (× 100)			
			PvZ	1-6	1 + 2	3	4-6	PvZ	1+2	3	4-6
			1-PvZ	1-6	1-6	1-6	1-6	1-PvZ	1-6	1-6	1-6
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Insectivora											
0575	<i>Solenodon paradoxus</i>	2	27.1	437	222	83.2	132	5.83	50.9	19.0	30.1
0577	<i>Tenrec ecaudatus</i>	3	25.7	288	129	57.0	102	8.21	44.9	19.8	35.3
0579	<i>Selifer setosus</i>	2	6.12	203	71.9	50.5	80.6	2.93	35.4	24.9	39.7
0581	<i>Hemicentetes semispin.</i>	1	3.26	90.3	33.6	22.4	34.3	3.49	37.2	24.8	38.0
0585	<i>Echinops telfairi</i>	2	1.26	70.0	26.9	17.1	26.0	1.77	38.5	24.4	37.1
0637	<i>Potamogale velox</i>	2	2.90	83.1	28.7	21.4	33.0	3.37	34.6	25.7	39.7
0689	<i>Erinaceus algirus</i>	3	22.0	263	119	48.3	95.2	7.71	45.4	18.4	36.2
0695	<i>Erinaceus europaeus</i>	5	25.2	307	137	60.2	109	7.61	44.8	19.6	35.6
0701	<i>Hemiechinus auritus</i>	3	12.6	163	63.6	37.7	61.2	7.17	39.1	23.2	37.7
0715	<i>Sorex araneus</i>	5	0.138	15.5	4.45	4.37	6.72	0.88	28.6	28.1	43.3
0769	<i>Sorex minutus</i>	1	0.064	8.58	2.64	2.64	3.30	0.74	30.8	30.7	38.5
0943	<i>Crocodyria flavescens</i>	2	0.635	36.9	12.5	10.5	13.9	1.69	33.9	28.5	37.6
1071	<i>Crocodyria russula</i>	2	0.137	16.2	5.64	4.64	5.90	0.84	34.9	28.7	36.5
1137	<i>Suncus murinus</i>	1	1.34	32.9	11.1	9.48	12.3	3.92	33.7	28.8	37.5
Scandentia											
1263	<i>Tupaia glis</i>	20	3.70	128	46.9	29.9	50.8	2.82	36.8	23.4	39.8
1269	<i>Tupaia minor</i>	2	2.58	91.8	29.6	22.7	39.5	2.73	32.3	24.7	43.0
1289	<i>Urogale everetti</i>	1	5.89	178	72.5	39.6	66.3	3.19	40.6	22.2	37.2
Primates											
Prosimians											
3199	<i>Microcebus murinus</i>	6	0.641	42.4	16.3	11.0	15.1	1.49	38.4	26.0	35.6
3205	<i>Cheirogaleus medius</i>	2	2.09	99.9	36.1	28.9	34.9	2.05	36.1	28.9	35.0
3215	<i>Lemur albifrons</i>	3	5.67	201	69.7	50.1	81.3	2.75	34.7	24.9	40.4
3239	<i>Lepilemur ruficaudatus</i>	3	3.48	109	37.8	30.4	41.1	3.08	34.6	27.8	37.6
3244	<i>Avahi laniger occident.</i>	2	1.30	68.8	27.1	16.5	25.2	1.86	39.4	23.9	36.7
3249	<i>Indri indri</i>	2	7.34	160	70.1	38.4	51.7	4.38	43.7	24.0	32.3
3251	<i>Daubentonia madagascar.</i>	1	56.3	629	268	129	232	8.22	42.6	20.5	36.9
3275	<i>Galagoides demidoff</i>	2	1.43	81.7	32.7	22.3	26.7	1.72	40.0	27.3	32.7
3282	<i>Tarsius spp.</i>	2	0.277	18.5	7.55	5.07	5.92	1.47	40.7	27.3	31.9
Simians											
3287	<i>Callithrix jacchus</i>	3	0.454	22.2	8.98	5.34	7.84	2.01	40.5	24.1	35.4
3325	<i>Aotus trivirgatus</i>	3	0.834	60.2	18.8	15.4	26.0	1.37	31.2	25.6	43.2
3371	<i>Ateles geoffroyi</i>	1	1.72	88.7	42.9	19.8	26.0	1.90	48.4	22.3	29.3
3453	<i>Cercopithecus mitis</i>	1	1.36	115	57.0	22.7	35.6	1.17	49.4	19.7	30.8
3477	<i>Colobus badius</i>	2	0.830	50.5	24.5	10.0	16.0	1.62	48.5	19.8	31.7
3549	<i>Pan troglodytes</i>	1	5.74	251	129	44.2	77.8	2.23	51.5	17.6	30.9
Macroscelidea											
7993	<i>Elephantulus fuscipes</i>	2	1.70	61.6	25.5	16.4	19.7	2.69	41.4	26.6	32.0
8011	<i>Rhynchocyon cirnei</i>	2	38.1	388	175	78.6	134	8.94	45.2	20.3	34.6
Means											
Tenrecinae	0577-0585	4						4.10	39.0	23.5	37.5
Erinaceidae	0689-0701	3						7.50	43.1	20.4	36.5
Soricidae	0715-1137	5						1.61	32.4	29.0	38.6
Basal Insectivora		12						3.91	37.3	25.0	37.7
Insectivora	0575-1137	14						4.01	38.0	24.6	37.4
Scandentia	1263-1289	3						2.91	36.6	23.4	40.0
Prosimians	3199-3282	9						3.00	38.9	25.6	35.5
Simians	3287-3549	6						1.72	44.9	21.5	33.6
Simians except <i>Aotus</i>		5						1.79	47.7	20.7	31.6
Primates	3199-3549	15						2.49	41.3	24.0	34.7
Macroscelidea	7993-8011	2						5.82	43.3	23.4	33.3
Overall	0575-8011	34						3.35	39.7	24.1	36.2



Figs. 2-5. Size indices of the MOB: three non-primate orders, two primate suborders and man (Fig. 2), insectivoran order and its families or subfamilies (Fig. 3), prosimian suborder and its families (Fig. 4), simian suborder and its families or subfamilies (Fig. 5). The indices are numerical expressions of the distances from the thick line given in Fig. 1, and are given in their percentage form ($\times 100$). Each of the small horizontal bars represents the size index for a species. On the left, the range (vertical bars) and the mean (O) of various groups are given, while on the right the sequences of means for the various groups are scaled. Water adapted Insectivora (Fig. 3) are marked by W; diurnal prosimians (Fig. 4) by D; nocturnal simians (Fig. 5) by N.

(range 11–16; $\bar{x}$ 13; n 5), and Cercopithecinae (range 5.5–17; $\bar{x}$ 12, n 7) in addition to *Aotus*; relatively low values were obtained in the Colobinae (range 1.2–5.8; $\bar{x}$ 3.1, n 3) as well as in *Homo*. Intermediate indices were found in the Cebidae (range 5.0 to 12; $\bar{x}$ 8.6; n 7; *Aotus* excluded) and in the Pongidae (range 5.6–10; $\bar{x}$ 7.8; n 3).

There is a definite trend in the size indices going from low Insectivora through prosimians and simians to man, the average indices being 100–64–10–3.6, respectively (Fig. 2).

MOB related to total brain and telencephalon: Because of the small size of the MOB in higher Primates, its relative size is better expressed as pro mille (‰) related to the total brain or telencephalon. The figures from the 76 species are given in Table 1 (columns 5 and 6) and averaged for several of the systematic groups. When expressed in this way clear trends from lower Insectivora via prosimians, simians, apes, and finally to man emerged.

The average pro mille values relative to the *total brain* were 108 in basal Insectivora, 69 in progressive Insectivora, 63 in Macroscelidea, 43 in Scandentia, 16 in prosimians, 1.4 in monkeys, and 0.6 in apes. The distinctly lowest pro mille value (0.09) was found in man, the highest (149) in *Setifer setosus* and in the Tenrecinae as a group (range 124–149; $\bar{x}$ 134; n 4). Thus in *Setifer* the MOB represents about one seventh of the total brain.

When examined in relation to the *telencephalon*, the pro mille portion of the MOB showed a still clearer decrease from lower Insectivora through man (Table 1, column 6). The average pro mille values were 177 in basal Insectivora, 113 in progressive Insectivora, 109 in Macroscelidea, 70 in Scandentia, 23 in prosimians, 18 in monkeys, and 0.75 in apes. The distinctly lowest pro mille value (0.11) was found in man, the highest (231) again in *Setifer setosus* and in the Tenrecinae as a group (range 212–231; $\bar{x}$ 220; n 4). Thus in *Setifer* and Tenrecinae between one fourth and one fifth of the telencephalon is MOB.

The given values depend, of course, not only on the size of the MOB, but to a great extent on the size of the total brain and telencephalon as the reference systems. In the higher Primates and in man the very low pro mille values were not only due to the low size of the MOB but also to the large brain and telencephalon.

When the MOB volumes were plotted against total brain and telencephalon volumes in double logarithmic scales, there was wide dispersal indicating the independence of the MOB from brain and telencephalon size. The correlation coefficients were very low (r^2 0.083 and 0.009) and the slopes of the regression lines were 0.137 and 0.0001.

Thus, when the telencephalon increases from low Insectivora through prosimians and simians to man, the volume size of the MOB does not change generally. For instance, man has a MOB volume which is only slightly larger than that of *Hemicentetes* (114 versus 93.6), but a telencephalon volume which is more than 2400 times larger.

Interspecific comparison of MOB components

Components related to total MOB: In the MOB there is a relatively small component, whose percentage size is highly variable in contrast to that of the other components. This is the deepest part of the MOB, which may be called the periventricular zone (PvZ). The percentage of this part relative to the total MOB varies from 0.74% in *Sorex minutus* to 8.94% in *Rhynchocyon cirnei* (1:12). The overall average is 3.35%, i.e. PvZ is on the average about 1/30 of the total MOB. The highest percentages were found in species which have both large absolute and relative MOB size, such as *Rhynchocyon*, *Solenodon*, *Hemichinus*, *Daubentonia*, *Erinaceus* and *Tenrec*. The lowest percentages were found in particularly small animals, such as shrews, which also have the absolutely smallest MOB volumes. The dependence of PvZ size on MOB size also appears from interstructural regression line analysis. When PvZ is plotted against MOB in a double logarithmic scale, the slope of the regression line is 1.538, which is distinctly higher than 1, indicating that in the increasing MOB, the PvZ increases much more than the rest of MOB. This inner zone seems to consist mainly of fibers (mostly glial?) which fill the room between the granular layer and the olfactory ventricle. This zone may not be a substantial component of the MOB in the narrow sense. Therefore, and because of its high variability, we have excluded it from the MOB complex for the calculation of its percentage composition.

The percentages of the three main components (layers 1 + 2, 3, and 4–6) show no clear trend from well developed to strongly reduced MOB. The composition of the MOB is relatively stable (see also ETIENNE 1970, 1972). The outer component (layers 1 + 2) varies from 30.8% in *Sorex minutus* to 51.5% in *Pan troglodytes* (1:1.7); the average is 39.7%. The main plexiform layer (layer 3) varies from 17.6% in *Pan* to 30.7% in *Sorex minutus* (1:1.8) with an average of 24.1%. Finally, the inner component (layers 4–6) varies from 29.3% in *Ateles* to 43.3% in *Sorex araneus* (1:1.5); the average is 36.2%.

Components related to body weight (size indices): After excluding PvZ, the size indices of the MOB (layers 1–6) were recalculated and it was found that the relatively small PvZ has only a limited effect on

the indices. In most cases, the indices with and without PvZ were identical or very similar. The largest differences were found for *Rhynchocyon* (220 with and 209 without PvZ) and *Daubentonia* (131 versus 125). Furthermore, the size indices of the three main components were similar to those of the total MOB as given in Table 1, column 4, and are therefore not presented in detail. There are, however, some special features which need attention.

First, in most species with relatively large MOBs (*Rhynchocyon*, *Solenodon*, *Tenrec*, Erinaceidae, *Daubentonia*) the highest indices were noted in layers 1 + 2. This was also reflected by the high slope of 1.082 (distinctly larger than 1), when layers 1 + 2 were plotted against layers 1–6 (MOB minus PvZ). There are, however, also some species with well developed MOB, in which other components have the highest indices: in *Galagoides*, layer 3 has the highest index, in *Selifer* and *Tupaia* spp. layers 4–6 are the highest.

Second, in all shrews, the size indices of layer 3 are markedly larger than those of the total MOB. This is mainly at the cost of layers 1 + 2. In hedgehogs, the relations are reversed. This results also from the average percentage values: layer 3 is relatively largest in the shrews (29.0%) but relatively smallest in the hedgehogs (20.4%) (see Table 2, column 9, means).

Discussion

(1) Methods of comparison

When plotting MOB versus body weight across all 76 species of the 4 orders (see stippled line in Fig. 1), the very low slope (0.188) might seem to indicate that the MOB is nearly independent of body size. This, however, is not true, since it can easily be read from Figure 1 in conjunction with Table 1 that within families, the large species have distinctly larger MOB than the small species and that this fact accounts for both insectivoran and primate families. The stippled line in Figure 1 results from the strong reduction of the MOB in higher Primates, which in general have larger body weights than Insectivora. When instead of Primates the still larger elephants (MOB data taken from MARSCHNER, 1970) are compared with Insectivora, the Insectivora-Proboscidea regression line coincides with the thick reference line of Figure 1, since elephants are similar to Insectivora in development of olfaction.

In previous papers (FRAHM et al., 1982; STEPHAN et al., 1981b, 1982) we have expressed reservations against the application of regression methods across orders when plotting brain or brain structures versus body weight, as practised by many authors (e.g.

BONIN, 1937; JERISON, 1973). It is important to carefully establish the slope of the reference line based on *narrowly related groups*, which are mainly families, but usually also suborders. The MOB is another good example that substantiates the validity of this restriction. Given this consideration, the regression method and the resulting size indices are irreplaceable for interspecific comparisons. When comparing the various components within a laminated structure (such as the MOB) the percentage values of these components may be more informative than their size indices; thus these values were presented and discussed in more detail.

(2) Laminar differentiation

Periventricular zone (PvZ): A large PvZ is associated with a large MOB, but it exists only in species whose heads (skulls) allow for a free expansion of the MOB. Additional characteristics appearing together in such cases are: (1) large olfactory ventricles and (2) more or less round shape of the MOB in cross sections. Species belonging to this type (percentage of PvZ relative to MOB in brackets) are: *Rhynchocyon* (8.94), *Daubentonia* (8.22), *Tenrec* (8.21), *Erinaceus* (7.71 and 7.61), *Hemiechinus* (7.17), and *Solenodon* (5.83). In the case of relatively large MOB but limited space all the characteristics described above are absent. Instead, the olfactory ventricle is missed, the shape in cross-sections is more oval or flat, and the proportion of PvZ is low, as e.g. in *Galagoides* (1.72). ETIENNE (1970) found no olfactory ventricle in *Galagoides* and found especially large ones in *Rhynchocyon*, *Daubentonia*, *Erinaceus* and *Tenrec*.

Laminar components (layers 1 + 2, 3, and 4–6): Within the simians *Aotus trivirgatus* has a MOB composition which differs from that of all other simian species by having a distinctly lower percentage of layers 1 + 2 and a distinctly higher percentage of layers 4–6. *Aotus* is the only nocturnal species in the otherwise diurnal suborder. Whether the special qualities of its MOB composition are related to its nocturnal activity might be confirmed in prosimians, a suborder in which most species are nocturnal, but several are diurnal.

The MOB components were measured in two diurnal prosimians (*Lemur*, *Indri*) and showed no significant differences from the nocturnal species (n 7). Thus the MOB composition in the prosimian suborder does not support speculations about a possible relation of nocturnal activity and the deviant MOB composition in *Aotus*.

The diurnal simians (i.e. *Aotus* excluded) have the highest average percentage in layers 1 + 2 and the lowest in layers 4–6, when compared with all the

groups so far investigated (see Table 2, means). This is in accordance with the general observations (obtained from light-microscopy) that the granular layer (layer 6) is especially reduced in higher Primates (e.g. STEPHAN, 1975, p. 229).

(3) Eco-ethological relations to MOB size

Feeding behaviour

In contrast to the findings on the accessory olfactory bulb (AOB) (STEPHAN et al. 1982), most of the size variation of the MOB among Primates, especially among prosimians, can be adequately explained by differences in feeding behaviour as characterized by diet preferences, feeding strategies and foraging techniques. With respect to diet preferences, HLADIK (1975, 1979) distinguishes three grades which differ from each other in the preferred food type and the proportional composition of various food supplements: Insectivorous and carnivorous forms which supplement their diet with various amounts of fruits and gum (grade 1); frugivorous species which also include the preferential gum- and seed-eaters that eat animal or vegetable diet as a protein supplement (grade 2); and folivores obtaining their protein mainly from leaves, shoots or flowers with a supplement of various amounts of fruit (grade 3). The proposed classification of the dietary specializations of the species used in this study were based on data published by NAPIER and NAPIER (1967), HLADIK (1975, 1979), KAY (1975), and HILL (1953, 1955, 1960, 1962, 1966). Further references are given by these authors.

The species investigated in this paper may be classified as follows:

Grade 1: *Microcebus murinus*, *Daubentonia madagascariensis*, *Loris tardigradus*, *Nycticebus coucang*, *Galagoides demidoffi*, *Tarsius spp.*

Grade 2: *Cheirogaleus major*, *C. medius*, *Perodicticus potto*, *Galago senegalensis*, *Otolemur crassicaudatus*, Callitrichidae, Cebidae except *Alouatta* Cercopithecinae except *Erythrocebus patas*, and among the Pongidae *Hylobates lar* and *Pan troglodytes*.

Grade 3: Lemuridae, Indriidae, *Alouatta seniculus*, *Erythrocebus patas*, Colobinae and *Gorilla*.

The cases of *Cercopithecus mitis* and *C. ascanius* are not clear. According to NAPIER and NAPIER (1967) and HYLANDER (1975), they include a large amount of leaves and bamboo shoots in their diet and thus also may be classified with grade 3.

Prosimians: Among prosimians (Fig. 4), the MOB indices vary widely (20–131; $\bar{x}$ 64; n 18). With re-

gard to the various feeding categories, the lowest mean index is found for grade 3, i.e. for the folivorous Lemuridae and Indriidae ($\bar{x}$ 38; range 20–68, n 7). However, folivorous Indriidae including *Indri*, a very specialized leaf-eater at the bottom of the scale, have a much lower mean index (24.5) than folivorous Lemuridae (63.5). Most of the folivores are diurnal species and locate food mainly by vision. One would expect that nocturnal species which have the same diet as the diurnal forms should compensate for the insufficient visual acuity by a greater development of the olfactory system. The folivorous *Lepilemur*, however, despite its nocturnal life habits, has a surprisingly low index, which is even lower than those of the day-active Lemuridae. *Avahi laniger*, the other nocturnal leaf-eater, has a very low MOB index similar to the diurnal Indriidae. *Lepilemur* differ from other leaf-eating species by their particular feeding strategy. Their staple food consists of the tough foliage of various trees commonly found in the forest (CHARLES-DOMINIQUE and HLADIK, 1971; HLADIK, 1979). We could not find any report on the exact feeding strategy of *Avahi laniger*. It is possible that, like *Lepilemur*, they feed on items which are sufficiently distinctive and ubiquitous to be easily found and recognized visually at a reasonable distance at night.

The relative size of the MOB among fruit and gum eating prosimians (grade 2) range from 63 in *Otolemur* to 99 in *Perodicticus* (average 83; n 5). On the average the MOB of this grade is better developed than that of the folivores (83 versus 38). All species belonging to this category are nocturnal forms. Since food sources are widely dispersed or have a patchy distribution, animals must travel over considerable distances. For instance, the distance covered by *Galago senegalensis* each night is approximately 2 km (CHARLES-DOMINIQUE and BEARDER, 1979). Locomotion and orientation in a complex environment certainly require efficient nocturnal vision. In fact scotopic vision is improved by a reflective tapetum (PARIENTE, 1979). Sight seems to be especially important in learning about features of the environment (CHARLES-DOMINIQUE, 1971; CHARLES-DOMINIQUE and BEARDER, 1979). However, there is sufficient evidence that for prosimians of grade 2, olfaction is the most important aid to the localization of food sources (CHARLES-DOMINIQUE and BEARDER, 1979). This is particularly true of those species which eat gums (PETTER et al., 1971). According to reports by CHARLES-DOMINIQUE and BEARDER (1979), bushbabies are attracted by the distinctive odour of fruiting or flowering trees or those bearing gum. This observation was confirmed by experimental evidence.

The greatest range of variation was found for the

insectivorous and carnivorous forms (*grade 1*). The MOB indices vary from 21 in *Tarsius*, which is close to the lowest index found for prosimians, to 131 in *Daubentonia*, which was the highest index found for all the Primates so far investigated. This marked variation in MOB size corresponds to a variation in the feeding strategies adopted by the different species.

Tarsius feed exclusively on animal prey comprising invertebrates, such as Orthoptera and spiders, and vertebrates such as small lizards and birds (HLADIK, 1979; NIEMITZ, 1979). According to observations made by NIEMITZ (1979) tarsiers locate their prey mainly by sound and sight. Smell seems to play a minor role if any at all. The importance of vision has already been confirmed by the high progression of the lateral geniculate body and the area striata (STEPHAN, 1969; BARON, 1981; STEPHAN et al., 1984; FRAHM et al., 1984).

The next group comprises *Loris tardigradus* (index 56) and *Nycticebus coucang* (index 62). Both utilize prey, including insects and other invertebrates which are generally avoided by other sympatric mammals and birds because of their repugnant smell and taste (HLADIK, 1979; ELLIOT and ELLIOT, 1967). The repulsive smell that is normally utilized by numerous invertebrate species as a defence mechanism acts as an attractant to these species of lorises. The generally high intensity of the normally repulsive odours may explain why the bulbi are less developed in *Loris* and *Nycticebus* than in other prosimians which hunt insects by smell. The third representative of the subfamily Lorisinae, *Perodicticus*, has a relatively large MOB (index 99). This may compensate the low development of the visual structures in this species (see STEPHAN et al., 1984; FRAHM et al., 1984).

Microcebus murinus (index 78) and *Galagoides demidoff* (index 120) resemble each other in terms of diet, feeding strategy and ecology (HLADIK, 1979). They feed mainly on insects supplemented by various amounts of birds, fruits and gums (CHARLES-DOMINIQUE, 1974, 1977). According to HLADIK (1979) hearing and sound detection of movements seem to be more important than olfaction in the predatory strategy of these species. Our results, however, indicate that smell too certainly appears to play a major role in prey detection. This seems to be especially true in the case of *Galagoides demidoff* whose MOB has a relative size similar to that found in terrestrial Insectivora.

The best developed MOB was found in *Daubentonia* (index 131) (Fig. 4), which exhibit a very specialized feeding strategy enabling them to feed on larvae of wood-boring insects. They locate the prey by smelling and hearing the insects beneath the bark (CARTMILL, 1974; HLADIK, 1979).

Simians: The olfactory system in simians (Fig. 5) differs from that found in prosimians in several respects. According to the size indices the MOB is on the average much less developed among the former (9.9; range 1.2–23; n 27) than among the latter (64, see above). The differential development of the sensory centers, particularly of the MOB, between prosimians and simians suggests that feeding strategies and foraging techniques may be different despite the fact that the same type of food is eaten by many species of the two taxonomic groups, and consequently, despite their high degree of convergence. In fact, among simians, visual signals assume an increasingly important role relative to olfactory cues. Improvement in vision has favoured improvement in food detection, foraging techniques and especially food manipulation. Prosimians execute hand movements of a fairly fixed pattern which limits the variability of food manipulation. Simians execute much more versatile finger movements which would be impossible without good vision.

The nocturnal New World monkey *Aotus* (marked N in Fig. 5) has the best developed MOB of all the simians. The lack of adequate ecological data about the night monkey makes comparisons with other species difficult. Its diet seems to include a great variety of food items, such as fruits, nuts, leaves, bark, gum, insects and small vertebrates (HLADIK and HLADIK, 1969; HILL, 1960). Despite its extraordinary scotopic adaptation which enables it to catch even insects in very dim light (SANDERSON and STEINBACHER, 1957), smell certainly plays a more important role than in any other simian primate.

Among the diurnal neotropical simians, the Callitrichidae generally have bigger MOBs (mean index 13) than the Cebidae (mean index 8.6 without *Aotus*). *Alouatta* which is predominantly a leaf eater (HLADIK, 1975) has, as might be expected, the lowest index (5.0) among Cebidae. The two subfamilies of paleotropical monkeys (Cercopithecidae) differ in their MOB development. The mainly folivorous Colobinae have the lowest mean index of all non-human primates (3.1). Among Cercopithecinae (mean index 11.9) the more folivorous species, *Cercopithecus ascanius* and *C. mitis*, however, do not differ from the mainly frugivorous species in terms of their MOB development. The only exception is the terrestrial grass-eating *Erythrocebus patas* (HYLANDER, 1975), which has an index (5.5) similar to that found in *Colobus* (5.8).

Finally, in Pongidae, no clear relation between size of the MOB and dietary adaptations was found. The mean index (7.8) of the more frugivorous forms (*Pan* and *Hylobates*) is similar to the index of the more folivorous *Gorilla* (7.7).

The small relative size of the MOB and the lack of a clear relationship between MOB size and food preferences suggest that olfactory cues play only a minor role in the feeding behaviour of diurnal simians.

Insectivora: If HLADIK's three grade classification of dietary types, which he originally applied to Primates (1979), is extended to include Insectivora it is likely that all species belong to grade 1 even though the variety of particular food items and the exact proportion of the vegetarian diet do vary considerably from one species to the other. While all eat invertebrates or small vertebrates, the use of odour in detecting and examining prey probably differs in ways which depend upon the occupied ecological niche. Therefore, the wide range of variation of the MOB size indices going from 38 in *Potamogale* to 170 in *Solenodon* (Fig. 3), which indicate great differences in olfactory capacities, should be interpreted as adaptations to different niches posing different ecological problems rather than as adaptations solely to the diversity and type of diet.

STEPHAN and SPATZ (1962) proposed a grouping of Insectivora based on a relationship between morphological features of the brain and eco-ethological adaptations. Using a slightly modified version of this classification, one can find a marked correspondance between the relative size of the MOB and the occupied niche. This correspondance is particularly relevant in the context of the variety of foraging strategies since a fundamental aspect of the ecological niche, besides the characteristics and distribution of food, is the method that the animal employs to exploit it.

Terrestrial or ground-dwelling Insectivora, including Solenodontidae, Erinaceidae, and many species of Tenrecidae, have the best developed MOB of all species of this order (Fig. 3). They shelter in stone crevices, tree hollows, ground holes, or dug-out burrows, but they forage on the ground. *Solenodon*, with the highest size index, hunts insects in plant debris and rotten logs using its claws, or by rooting in the soil with its snout (COCKRUM, 1962). Although the exact role of the various senses in foraging is not known, it is likely that an animal, when adopting the typical foraging pattern with its nose to the ground, is searching for olfactory cues.

Similar behaviour patterns are observed in tenrecs (EISENBERG and GOULD, 1970) and hedgehogs (HERTER, 1957; KINGDON, 1974) which are also terrestrial surface foragers. Foraging generally consists of probing with the nose in the mat of decaying vegetation or inserting the nose in crevices or between rotten logs; this is certainly an olfactory investigation.

Among the surface foraging Tenrecinae, differences in locomotor ability and in the degree of food specialization are not reflected by differences in the develop-

ment of the MOB: *Setifer* and *Echinops*, which are more or less semiarboreal climbers, differ in their MOB indices. Furthermore, both of them and also *Tenrec* tend to be more generalized omnivores. On the other hand, *Hemicentetes*, a preferential earthworm feeder, seems to be much more specialized in its foraging strategies (EISENBERG and GOULD, 1970). However, its MOB index is similar to that of *Tenrec*.

The three Oryzictinae of the genus *Microgale* are surface foragers with moderate or good climbing abilities. Their MOB indices are very similar and fall within the range of variation of the terrestrial Tenrecinae.

Perhaps the best known insectivoran surface foragers are the Erinaceidae. The size development of their MOB is similar to that of the other terrestrial Insectivora. Oddly, there is no agreement among authors about their olfactory capacity and the precise role of olfaction in foraging behaviour. According to HERTER (1957), hedgehogs move along with their noses only a few millimeters from the soil when foraging and produce audible sniffing sounds while inhaling and exhaling. Objects with strong odours can be detected from a distance of 11 meters. Furthermore, it has been shown that they ignore prey hidden behind a glass cover but can detect food buried three or four centimeters in the ground (HERTER, 1967). On the other hand, BRETTING (1972), who determined threshold levels of perception, considered the olfactory capacity of *Erinaceus europaeus* mediocre and comparable to that of man. Such comparisons may be misleading since they do not take into account a specialized sensitivity to odorants which may be biologically significant for one group but not for the other. This has been documented e.g. in WRIGHT's (1964) comparative analysis of the olfactory responses in dog and man.

With the exception of some semiaquatic forms, most of the Soricidae may be considered subterranean but not fossorial forms even if they occasionally dig burrows. They inhabit terrains covered by vegetation which offers good hiding places; they shelter under stones or in burrows of moles and small rodents. Subterranean shrews collect food by tunneling through the loose accumulation of plant detritus (RUDGE, 1968), in narrow spaces and crevices beneath rocks and logs, or by patrolling in surface runways and burrow systems of small mammals. We do not yet know to what extent olfactory cues are used by shrews in the detection of food. Observers disagree about the role of their sense of smell in finding food. While some authors maintain that shrews hunt almost entirely by smell (MIDDLETON, 1931; HOLLING, 1958; GRÜN-WALD, 1969), others agree that olfactory cues are of minor importance in detecting prey, and that food

seems to be located mainly by means of the extraordinarily sensitive long whiskers in the shrew's muzzle (BLOSSOM, 1932; HAMILTON, 1940; RUDGE, 1968; KINGDON, 1974). Hearing seems to be also important in hunting (SKARÉN, 1978). Although shrews can detect buried or openly available food items by olfaction (GRÜNWARD, 1969; VLÁSAK, 1970; SCHMIDT, 1979) and may distinguish the odours of different kinds of hidden prey, such as healthy and fungus-infected cocoons (HOLLING, 1958), it seems that shrews feed rather opportunistically on available prey encountered more or less accidentally during their regular excursions at various levels of the substratum. Our results indicate that subterranean shrews have less developed olfactory centers than Insectivora which hunt on the surface (Fig. 3). This supports the view that olfaction plays a minor role in the foraging strategy of Soricidae as compared with surface-foraging Insectivora.

Among true fossorial Insectivora, there are two types differing by their digging technique and by the relative size of their olfactory bulbs. The size indices of *Talpa* and *Oryzomys* are within the range of variation of the size indices of subterranean shrews, whereas those of the fossorial Chrysochloridae are similar to the indices of the ground foragers. Our knowledge about the feeding strategies of both types is rather poor.

It seems that talpids, during their regular excursions through complicated tunnel systems, which essentially serve as pitfall traps (MELLANBY, 1966), collect prey in a manner similar to subterranean shrews. One may therefore suppose that, as in shrews, olfactory cues do not play an important role in the detection of food among fossorial talpids. In fact, they ignore prey which are only a few centimeters away. Among Tenrecidae, *Oryzomys* is the most pronounced subterranean type. Like common moles, it possesses powerful digging claws and has similar MOB size indicating that, in contrast to terrestrial Tenrecidae, olfactory stimuli certainly play a less important role in the detection of food.

The second fossorial group, the Chrysochloridae, dig tunnels in soft soil and vegetable debris just under the surface of the ground as they search for prey, usually worms, which seem to be detected by smell. However, for food other than worms, Chrysochloridae rely on actual contact (KINGDON, 1974).

The most reduced olfactory bulbs, as emphasized in previous publications (STEPHAN and BAUCHOT, 1959; BAUCHOT and STEPHAN, 1968; STEPHAN and KUHN, 1982), are found in the semiaquatic forms (marked W in Fig. 3) regardless of their taxonomic affiliation (mean index 60). Since, in mammals, olfaction is linked with breathing, it is of no use in de-

tecting prey under water. Therefore, the reduced size of the MOB in those species that obtain a considerable portion of their food in the water is not surprising.

Macroscelidea: Both species of Macroscelidea used in this study differ from each other in the relative development of their MOB (Fig. 2). Those of *Elephantulus* have the same relative size as in ground-dwelling Insectivora. *Rhynchocyon*, on the other hand, has bulbs which are almost twice as large. In fact, it has by far the best developed MOB of all species investigated in this study. According to KINGDON (1974), *Rhynchocyon* seem to sniff continuously at their immediate surroundings with its probe-like nose. This certainly indicates that they rely on olfactory and tactile stimuli as the most important, if not the exclusive, means of finding food.

Scandentia: In the tree-shrews, which also belong to grade 1 of HLADIK's dietary types, the size indices of the olfactory bulbs have values similar to those of the terrestrial Insectivora (mean index 135) (Fig. 2). Although they are arboreal, they forage mainly on the surface of the forest floor (NAPIER and NAPIER, 1967). Olfactory stimuli seem to be a very important source of information in their feeding strategy.

Social behaviour

In addition to odours associated with food the environment comprises of odours emanating from a countless variety of sources with precise chemical characteristics. There is no doubt that for many species chemical stimuli constitute an important source of biologically meaningful information. However, the relative importance of odours as sources of information vary according to the ecological, morphological and physiological characteristics of the species. Perhaps the most important vectors of chemical information besides odours produced by food items are pheromones produced by conspecifics and allomones produced by individuals of other species.

It seems obvious to ask whether the amount and the diversity of chemical communication is parallel to the quantitative development of the olfactory system as reflected by the size indices of the MOB. The role of the accessory olfactory bulb (AOB) in social and particularly in sexual behaviour has been discussed in a previous paper (STEPHAN et al., 1982). It has been stressed that the involvement of two olfactory systems in social and sexual behaviour can be explained by differences in the processing mechanism for releaser and primer pheromones. The relative importance of olfactory signals may be deduced from the number of specialized scent-glands and the variety of pheromones produced, the variety and frequency of

scent-marking behaviour, the frequency of olfactory investigation, and the frequency of various types of nasal contact. However, it is possible that olfactory information can be transmitted without any observable behaviour and therefore our notions of marking behaviour and nasal contact may be overly simplistic.

Prosimians: All prosimians without exception possess various scent-glands and practise one or several forms of olfactory marking. Among this group, the nocturnal species that are solitary rely mainly on olfactory signals (SCHILLING, 1979). They have, with the exception of *Tarsius*, the best developed olfactory bulbs among Primates (mean index 86). Some diurnal species (e. g. *Lemur fulvus*, *Varecia variegata*, *Propithecus verreauxi*), which are in general gregarious, exhibit displays which have both visual and olfactory components (PETTER, 1962; JOLLY, 1966; EVANS and GOY, 1968; MERTL, 1976; SCHILLING, 1979). In contrast to the nocturnal solitary species, the size of their olfactory bulbs is reduced (mean index 51). Other species with still smaller olfactory bulbs like the diurnal *Indri* (index 20) and the nocturnal semisocial *Lepilemur* (PETTER, 1962) (index 41), do not practice scent marking very regularly nor do they exhibit frequently other olfactory behaviour patterns (POLLOCK, 1975; SCHILLING, 1979). A similar reduction of the size of the MOB was observed in both subspecies of *Avahi laniger* (mean index 26.5), which are nocturnal forms and gregarious in that they live in family groups (SCHILLING, 1979). No precise information about their marking behaviour is available. Based on the type of their social organization and the development of their main olfactory system, it is likely that olfactory communication is less important than among the solitary nocturnal forms. *Tarsius*, despite the relatively reduced size of its MOB as compared to that of other nocturnal prosimians, nevertheless exhibits various patterns of scent-marking (NIEMITZ, 1979). These observations suggest that in *Tarsius*, olfaction is an important channel of communication irrespective of its apparently minor role in feeding behaviour.

Simians: There seems to be general agreement based on morphological and behavioural evidence that among simians, South American monkeys have a more powerful olfactory communication channel than the Old World species. For instance, histomorphological investigations have shown that neotropical simians possess a greater variety of specialized odour-producing glands than do those of the Old World (WISLOCKI, 1930; EPPLE and LORENZ, 1967; EPPLE, 1974; EISENBERG, 1977; EPPLE and MOULTON, 1978). Furthermore, scent-marking has been reported for most of the neotropical simians (for ref. see OPPENHEIMER, 1977) but the paucity of reports on marking

behaviour in Old World simians indicates that the importance of olfactory signals in territorial marking is reduced in these species (GARTLAN and BRAIN, 1968; GAUTIER and GAUTIER, 1977). However, there are numerous reports of olfactory investigation of the genitalia by both males and females and to a lesser degree of other odour-producing parts of the body, especially in Cercopithecinae (for references and detailed discussion see GAUTIER and GAUTIER, 1977). In fact, Cercopithecinae have the best developed olfactory system among Old World simians. The mean index for their MOB (11.9) is similar to that found in the diurnal neotropical monkeys (10.4). One may therefore assume that, in addition to similar feeding strategies both groups (with the exception of the species already mentioned) resembles each other in the relative importance of the olfactory communication channel.

Scandentia: In *Tupaia*, the existence of scent glands and intensive marking, such as urine dribbling, have been reported by several authors (SPRANKEL, 1961; MARTIN, 1968; SPRANKEL and RICHARZ, 1976).

Insectivora: Most Insectivora possess various scent glands (for references see PODUSCHKA, 1977; BAXTER and MEESTER, 1980). According to THIESSEN (1977), animals with a high basal metabolic rate are able to produce a great quantity of glandular secretions. One may assume that, for Insectivora, which are known to have a high basal metabolism, the use of pheromones is a very economic means of communication. Furthermore, the particular characteristics of their terrestrial or subterranean ecotope, which favours a dispersed organization of the populations (NEVO, 1979), renders olfaction the most efficient channel for communication. In fact, marking behaviour has been observed in representatives of most of the insectivoran families and ecological types (PODUSCHKA, 1977; BAXTER and MEESTER, 1980).

Given the present state of our knowledge, we are not able to correlate the relative importance of the olfactory channel with the differential development of the olfactory bulbs among the various types of Insectivora. The available data, however, strongly suggest that, independently of its precise role in feeding behaviour, olfaction is the predominant sensory channel in intraspecific and interspecific interactions among Insectivora even in species with apparently reduced olfactory bulbs, such as semiaquatic forms.

General conclusions

The major trends in the differential development of the MOB that emerge from our results and the previous discussion may be summarized as follows:

1. There is a clear trend pointing to a constant reduction of MOB size from Insectivora to higher Pri-

mates, with *Homo* having a very poorly developed MOB. Scandentia and especially Macroscelidea have size indices which are similar or even higher than those of the terrestrial Insectivora.

2. Within each taxonomic group, the size development of the MOBs is largely linked to dietary adaptations: insectivorous or carnivorous species (grade 1) have respectively the best, and the folivorous (grade 3), the least developed MOBs. Frugivorous species occupy an intermediate position.

3. Within each dietary type, size differences of the MOB can be explained by the different characteristics of the occupied niche. Among Insectivora, ground-dwelling species occupy the upper, and semi-aquatic species, the lower positions on the size scale for the MOB. One may therefore assume that the particular size development of the MOB reflects diversity in the behavioural patterns associated with olfactory information in the feeding strategy of a particular species. In this respect, the cases of the two fossorial forms are particularly informative. The fossorial Talpidae, similar to subterranean shrews, are "incidental" gleaners in the sense that they collect opportunistically all prey they encounter on their regular patrol through the tunnel system which serves, as already mentioned, as a kind of pitfall. Chrysochloridae, on the other hand, are able to dig out worms they have detected by olfaction.

Among Primates, in closely related species, nocturnal forms have in general better developed MOBs than diurnal forms. But again, the involvement of olfaction in various aspects of feeding behaviour varies among the nocturnal species. For instance, despite their nocturnal life habits, *Lepilemur*, *Avahi*, and *Tarsius*, do not seem to rely primarily on olfaction when foraging.

4. Generally speaking, the importance of olfaction in feeding behaviour is paralleled by a similar importance of its role in social behaviour. The few but very significant exceptions in which olfaction constitutes an important if not the most important channel for communication despite its reduced role in trophic behaviour represent cases of a selective use of the environment which may be the result of a shift of the feeding niche. Otter-shrews and other semi-aquatic Insectivora are a typical example in this respect. As predators in limnetic ecosystems, they rely on the trigeminal system as the main sensory channel in the search for food (STEPHAN and KUHN, 1982); as solitary-dwelling animals of the terrestrial and subterranean ecological zone, however, they rely on olfaction as the most efficient sensory modality for communication. The tarsiers represent another interesting example. As mentioned above, they feed exclu-

sively on animal prey and are well adapted to catch quickly moving animals, such as flying insects (NIEMITZ, 1979). The optimal exploitation of such a food source requires good vision and hearing, and smell certainly is of minor importance. In this connection, it is interesting to note that the retina of *Tarsius* possesses an area centralis with a fovea (ROHEN and CASTENHOLZ, 1967). Olfactory signals, especially of a deferred nature, such as scent-marks, remain important for the nocturnal tarsiers with their pronounced territorial behaviour and predominant solitary life habits.

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